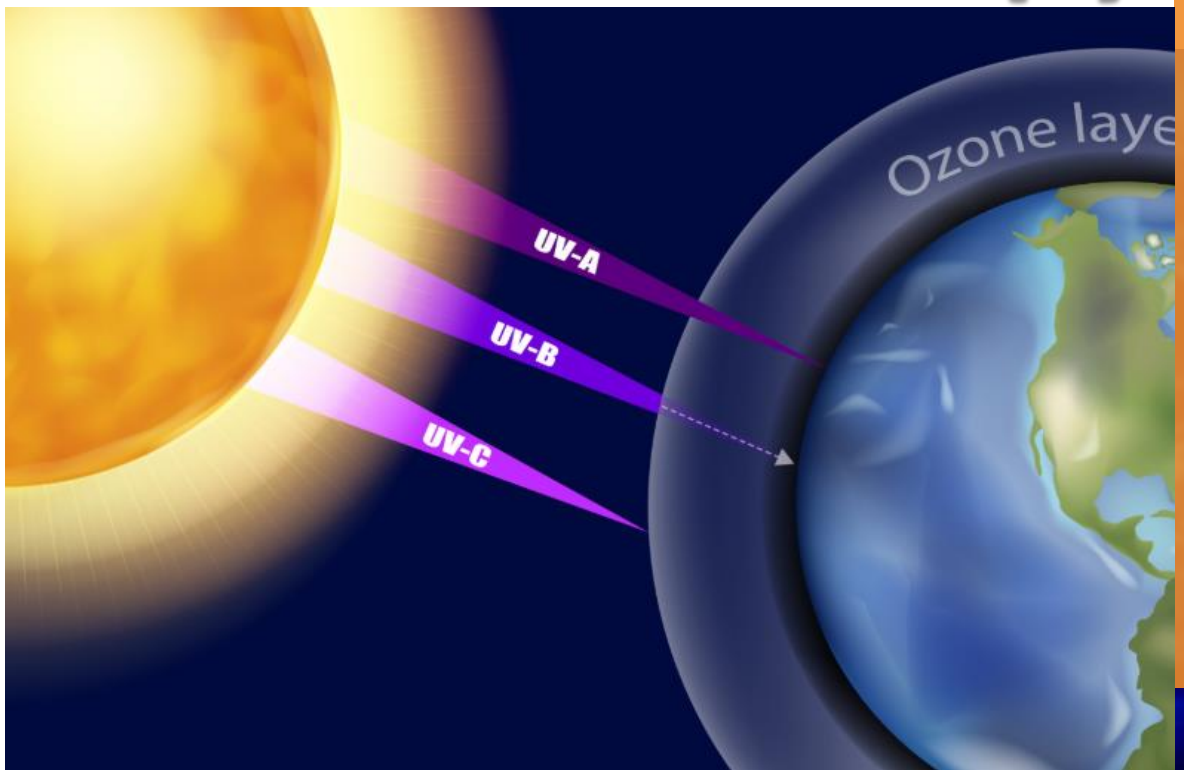
Ultraviolet Therapy



Prof. Dr. Mahmoud Ewidea,
Dean faculty of physical therapy,
Kaferelshek university.

Table Of Contents

Ultraviolet Therapy



01. INTRODUCTION

02. PRODUCTION

03. PHYSIOLOGICAL effects

04. THERAPEUTIC effects

06. INDICATION

07. CONTRAINDICATION,

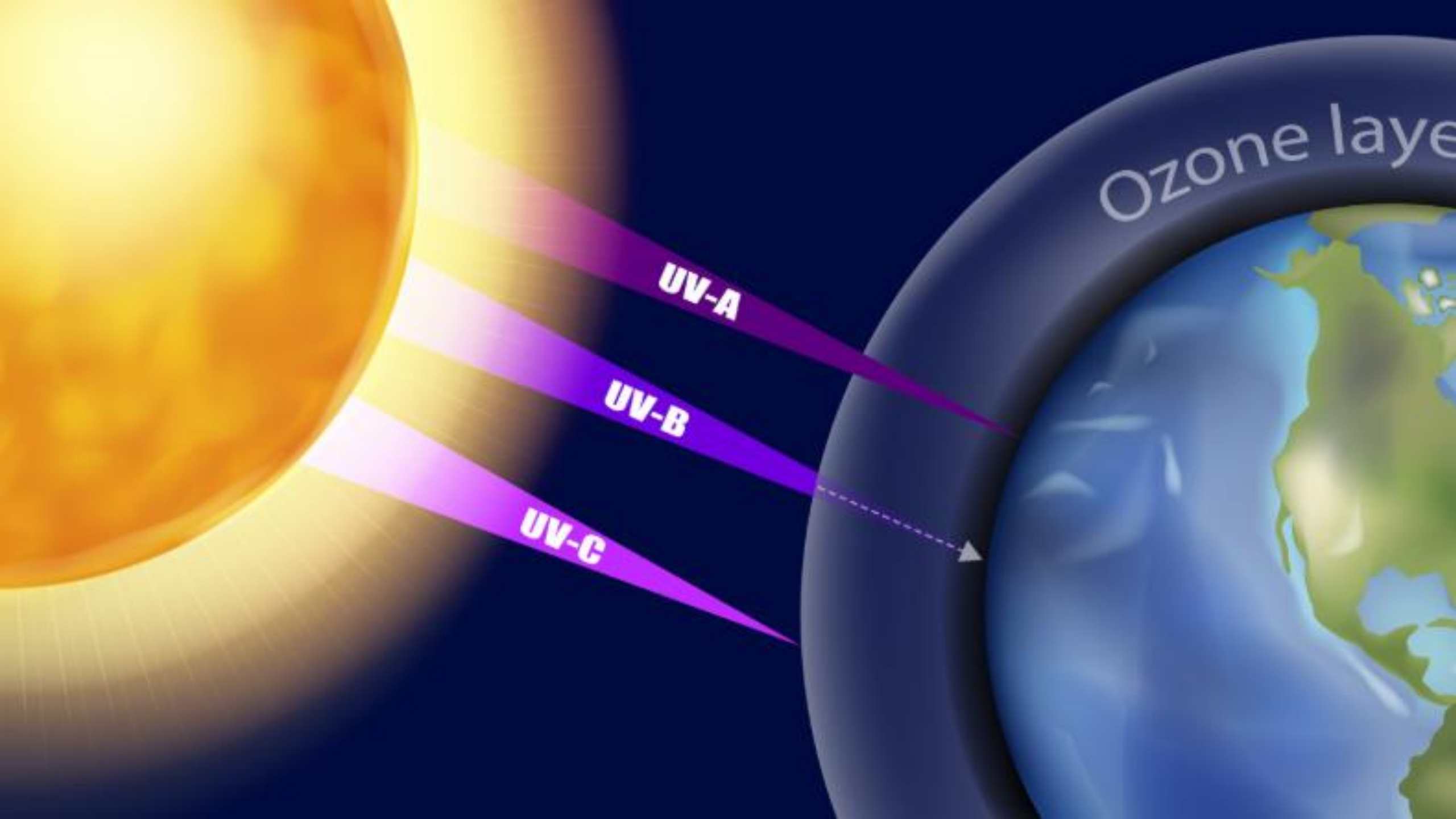
08. PRECAUTION & DANGERS

09. questiones

Ultraviolet Radiation (UVR)

- Ultraviolet radiation is electromagnetic energy, which is invisible to the human eye, with wavelengths between 100 nm to 400 nm. Divided into three ranges:
  - UV-A = Near UV (315nm-400nm)
    - Little or no physiologic effect
  - UV-B = Middle UV (280nm-315nm)
    - Sunburn and age-related skin changes
  - UV-C = Far UV (100nm-280nm)
    - Bactericidal




UV-A
UV-B
UV-C
Ozone laye

Ultraviolet radiation

- Ultraviolet radiation is produced by high-temperature surfaces, such as the Sun, in a continuous spectrum and by atomic excitation in a gaseous discharge tube as a separate spectrum of wavelengths. Most of the ultraviolet radiation in sunlight is absorbed by oxygen in Earth's atmosphere, which forms the ozone layer of the lower stratosphere. the ultraviolet that does reach Earth's surface, almost 99 percent is UVA radiation.

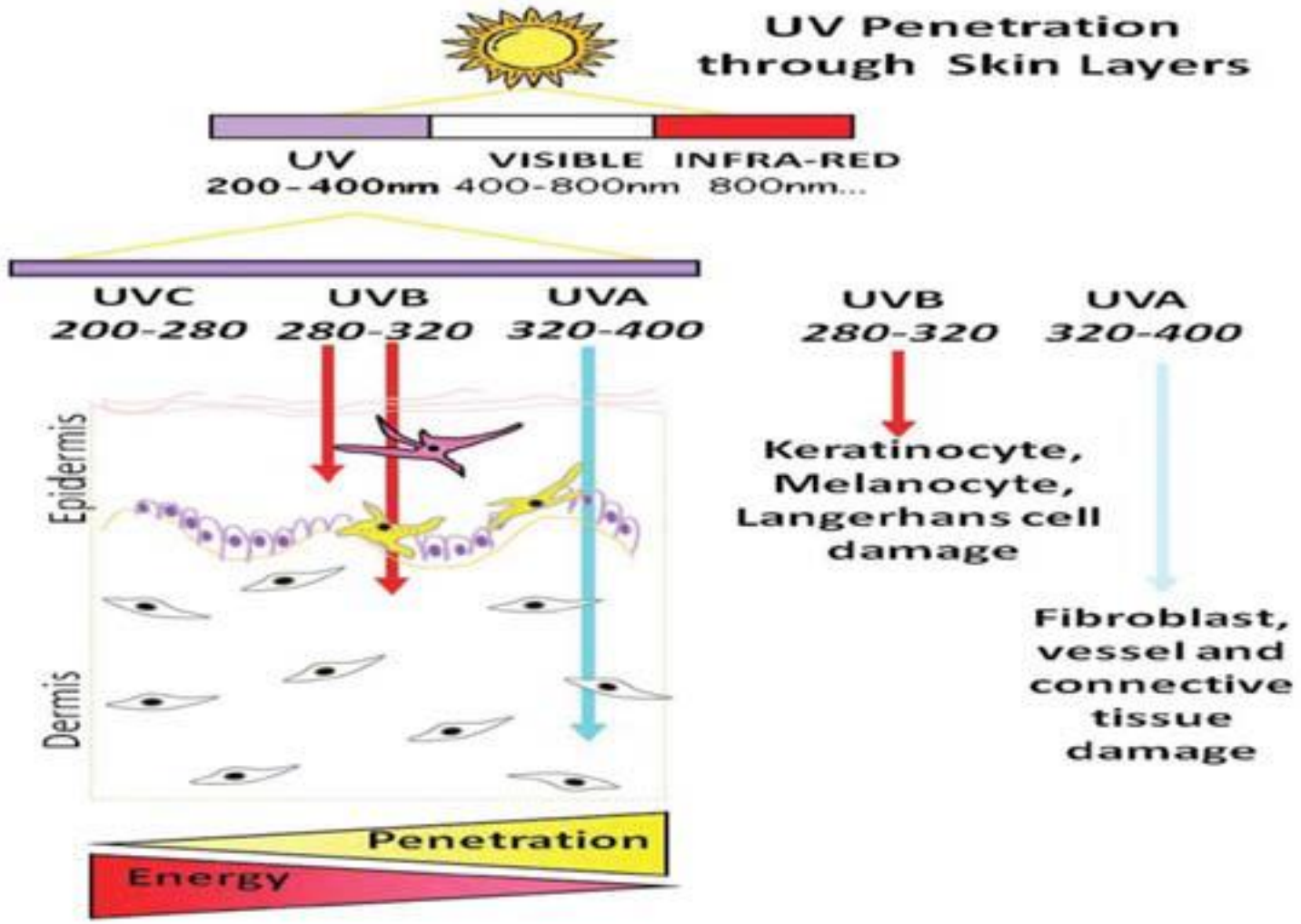


ultraviolet

- When the ozone layer becomes thin, however, more UVB radiation reaches Earth's surface and may have hazardous effects on organisms. For example, studies have shown that UVB radiation penetrates the ocean's surface and may be lethal to marine plankton to a depth of 30 metres (about 100 feet) in clear water.



Types of Ultraviolet



Ultraviolet Radiation

Most likely to be used

- UV-B or UV-C

UV depth of penetration is 1 to 2 mm

Physiologic effects are superficial

Used to treat various skin disorders

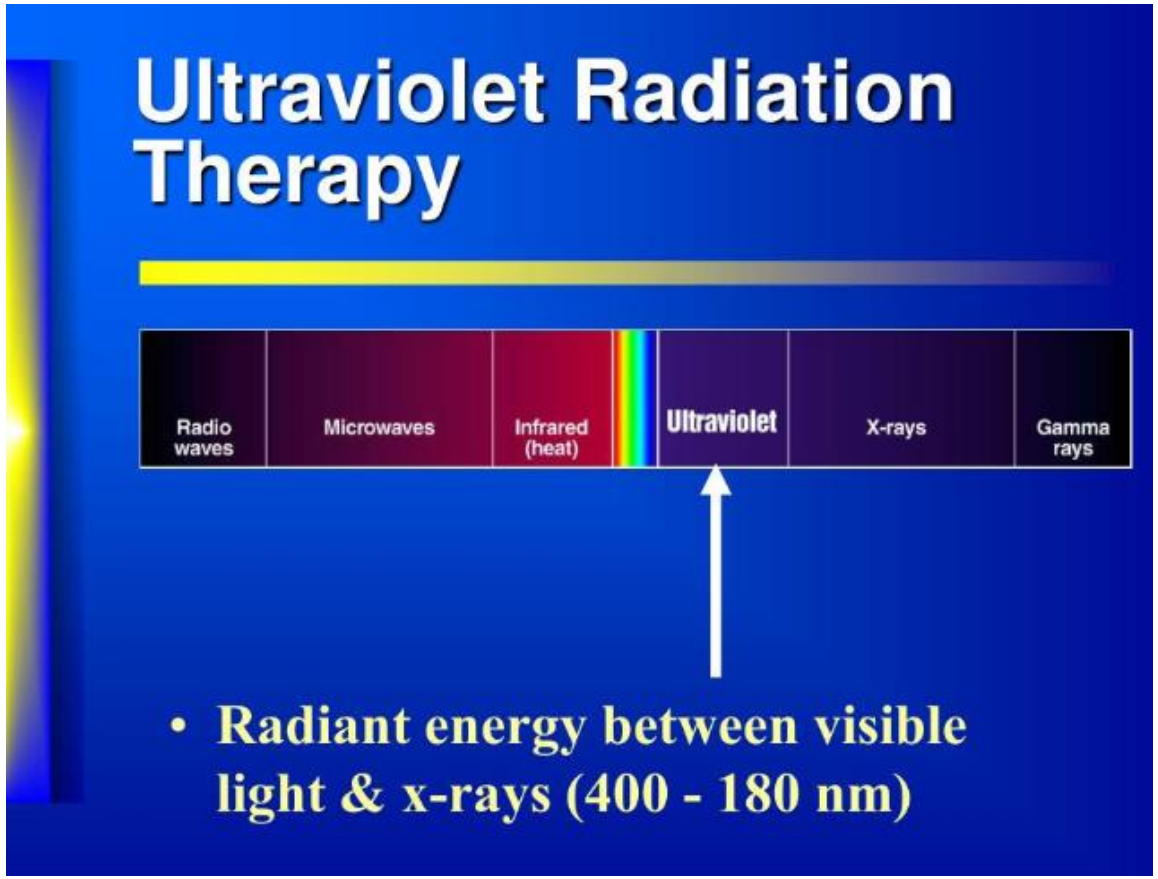
ozone and stratosphere screen out UVC and most of UVB from the sun's radiation.

ultraviolet radiation vs X-rays



ultraviolet radiation vs X-rays

- Unlike X-rays, ultraviolet radiation has a low power of penetration; hence, its direct effects on the human body are limited to the surface skin. The direct effects include reddening of the skin (sunburn), pigmentation development (suntan), aging, and carcinogenic changes. Ultraviolet sunburns can be mild, causing only redness and tenderness, or they can be so severe as to produce blisters, swelling, seepage of fluid, and sloughing of the outer skin.

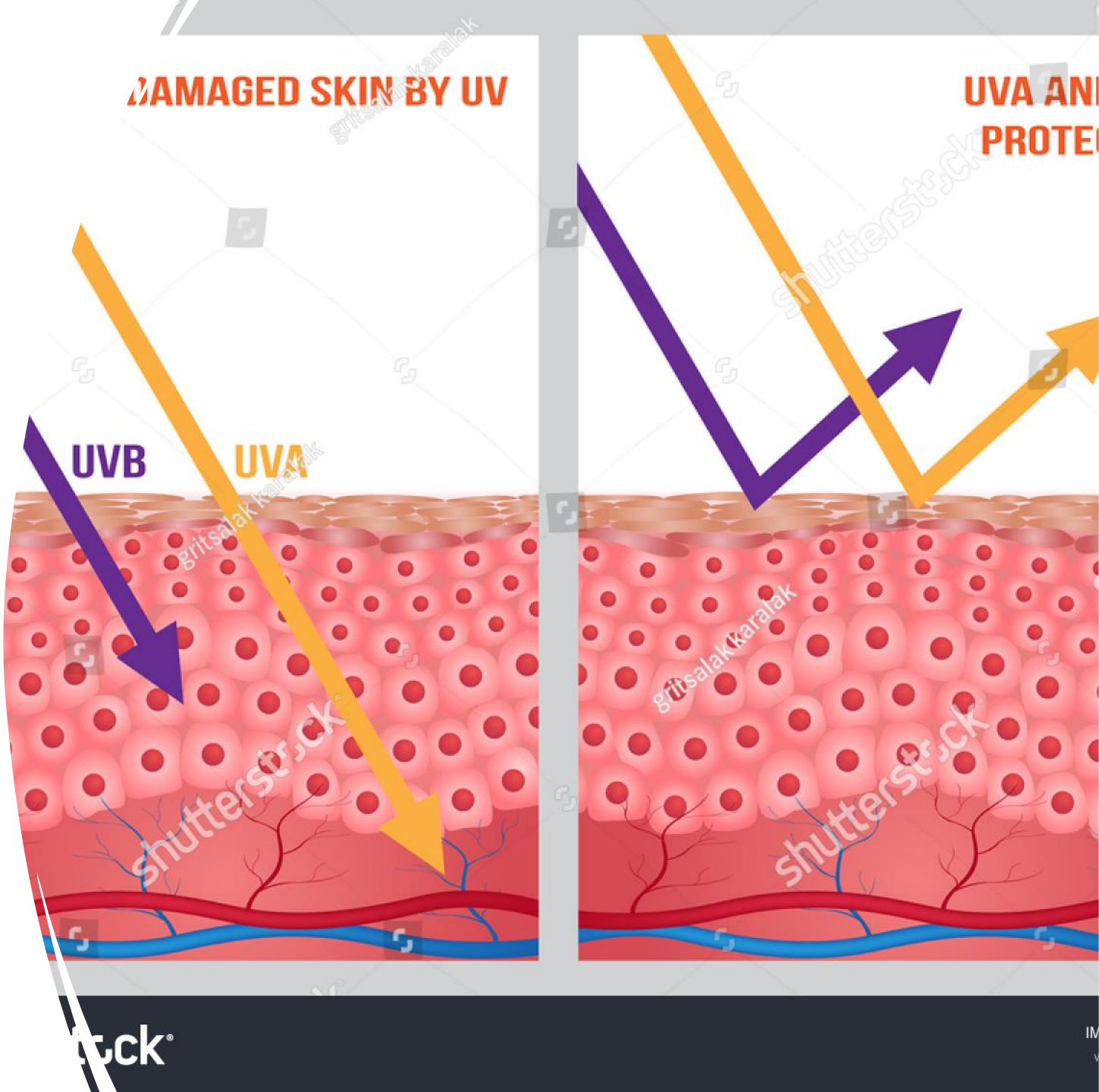




Production of Ultraviolet Radiations (UVR)

- Radiant sources, like sun, can produce UVR if the temperature is high enough However, it is usually produced by the passage of a current through an ionized vapour – often mercury vapors.
- !! Gases do not conduct current well at normal temperatures and pressures but can be made to do so at low pressure or high temperatures.

- The blood capillaries (minute vessels) in the skin dilate with aggregations of red and white blood cells to produce the red coloration. Tanning is a natural body defense relying on melanin to help protect the skin from further injury.

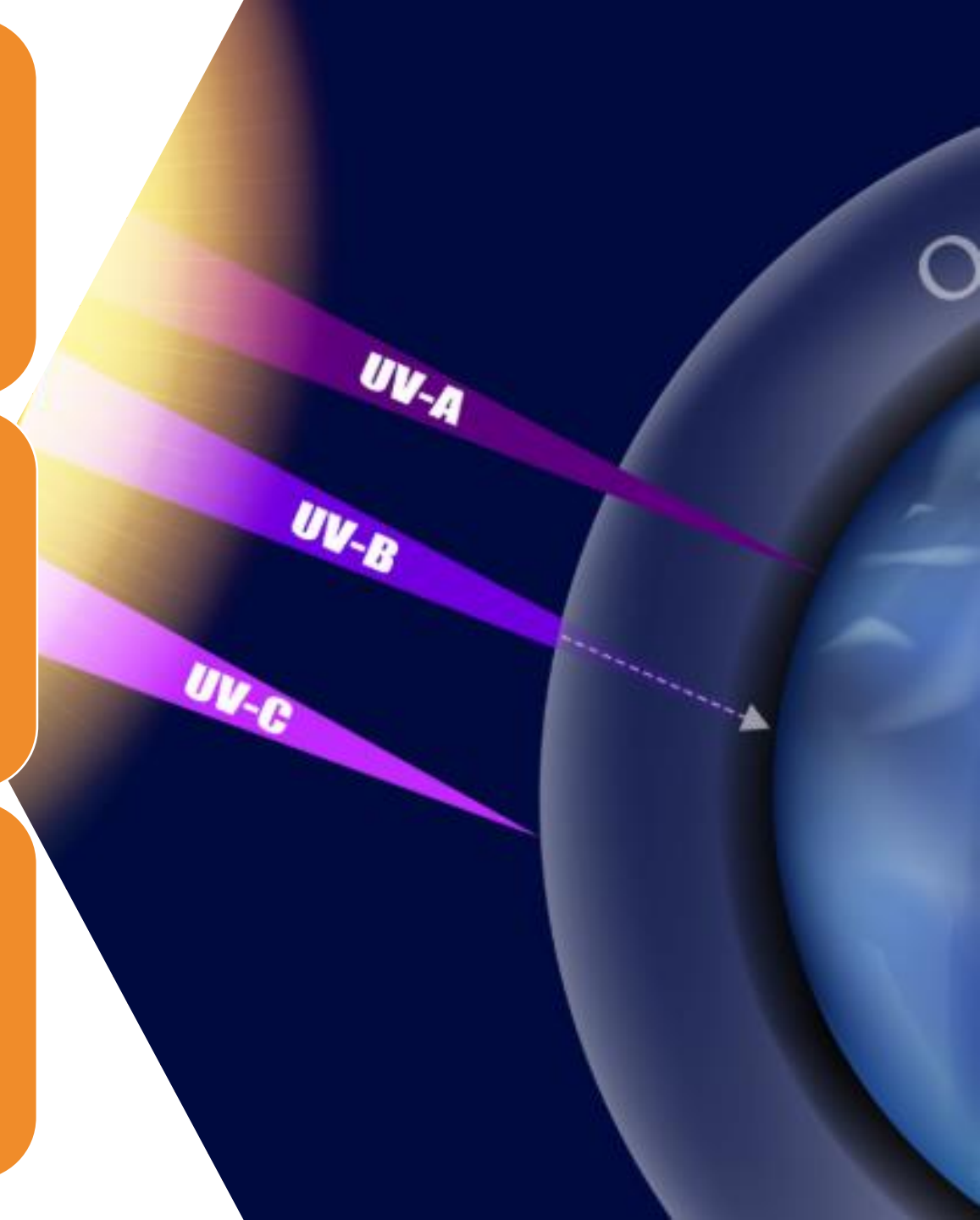


Why X-rays penetrate deeper than ultraviolet (UV) light?

primarily due to their higher energy and shorter wavelength. Here’s a more detailed explanation

Energy Levels: X-rays have much higher energy photons compared to UV light. This high energy allows X-rays to interact with matter at a deeper level, enabling them to break atomic bonds and pass through dense materials, including human tissues and various substances.

Wavelength: X-rays have shorter wavelengths than UV light. The shorter wavelength allows X-rays to interact less with the electrons in atoms. This means they can pass through materials more easily, as they are less likely to be absorbed or scattered.

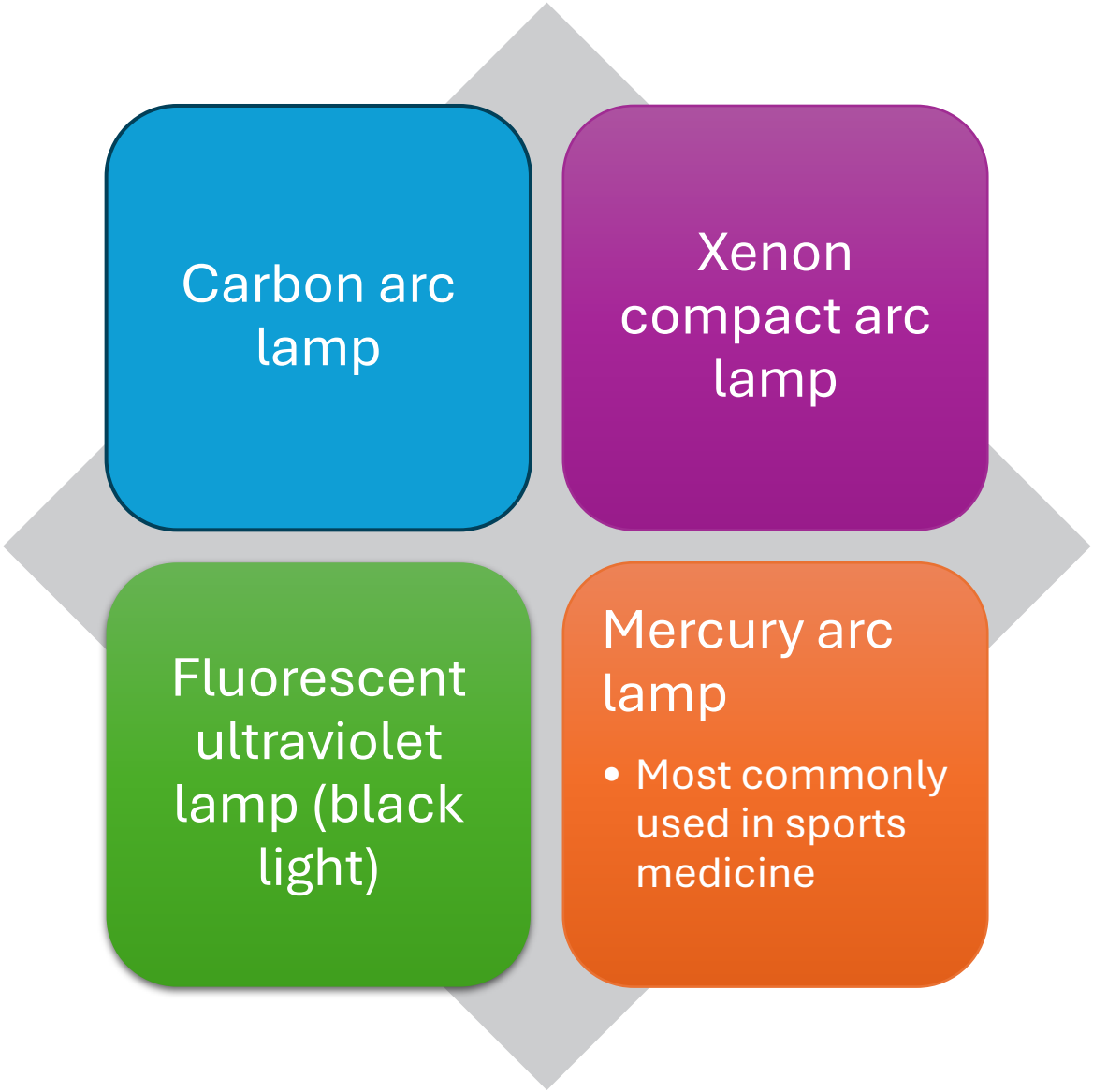


Cont.

Interaction with Matter: When light interacts with matter, it can be absorbed or scattered. UV light tends to be absorbed by electrons in atoms, especially in organic materials. X-rays, on the other hand, can penetrate deeper without being significantly absorbed, which is why they are used in medical imaging.

In summary, the combination of higher energy and shorter wavelength allows X-rays to penetrate deeper than ultraviolet light.

Ultraviolet Generators

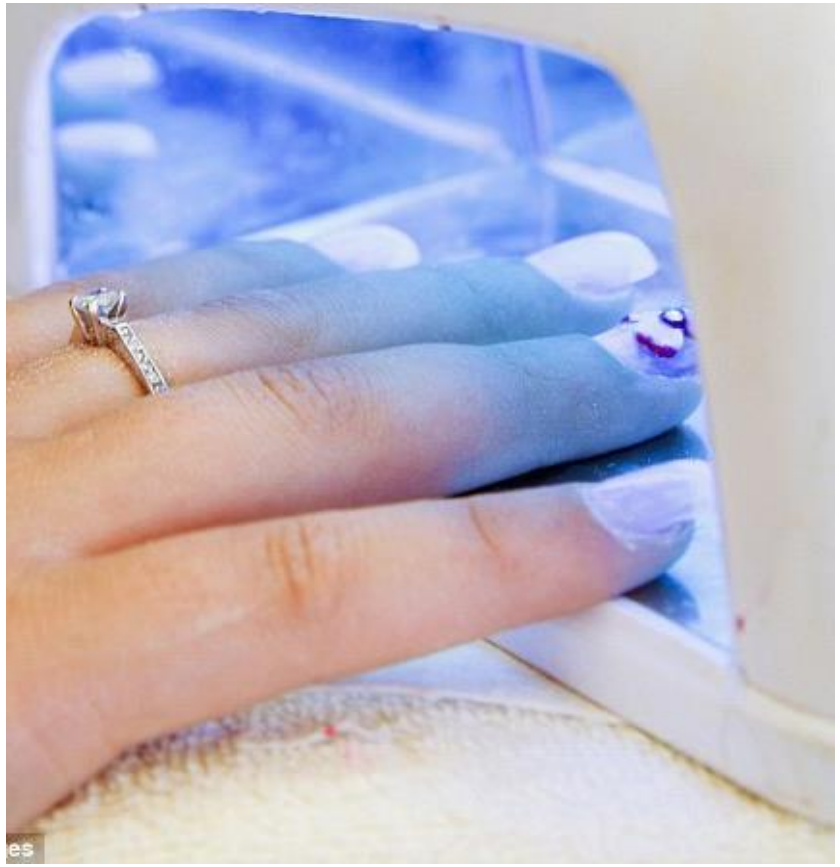


Skin vs ultraviolet

Melanin is a chemical pigment in the skin that absorbs ultraviolet radiation and limits its penetration into tissues. A suntan occurs when melanin pigments in cells in the deeper tissue portion of the skin are activated by ultraviolet radiation, and the cells migrate to the surface of the skin.

When these cells die, the pigmentation disappears. Persons of light appearance have less melanin pigment and so experience the harmful effects of ultraviolet radiation to a greater degree. The application of sunscreen to the skin can help to block absorption of ultraviolet radiation in such persons.

- Constant exposure to the Sun's ultraviolet radiation induces most of the skin changes commonly associated with aging, such as wrinkling, thickening, and changes in pigmentation. There is also a much higher frequency of skin cancer, particularly in persons with fair skin. The three basic skin cancers, basal- and squamous-cell carcinoma and melanoma, have been linked to long-term exposure to ultraviolet radiation and probably result from changes generated in the DNA of skin cells by ultraviolet rays.



Carbon arc lamp



Fluorescent ultraviolet lamp



Mercury arc lamp



UV Rays

How Do They Affect Your Skin?



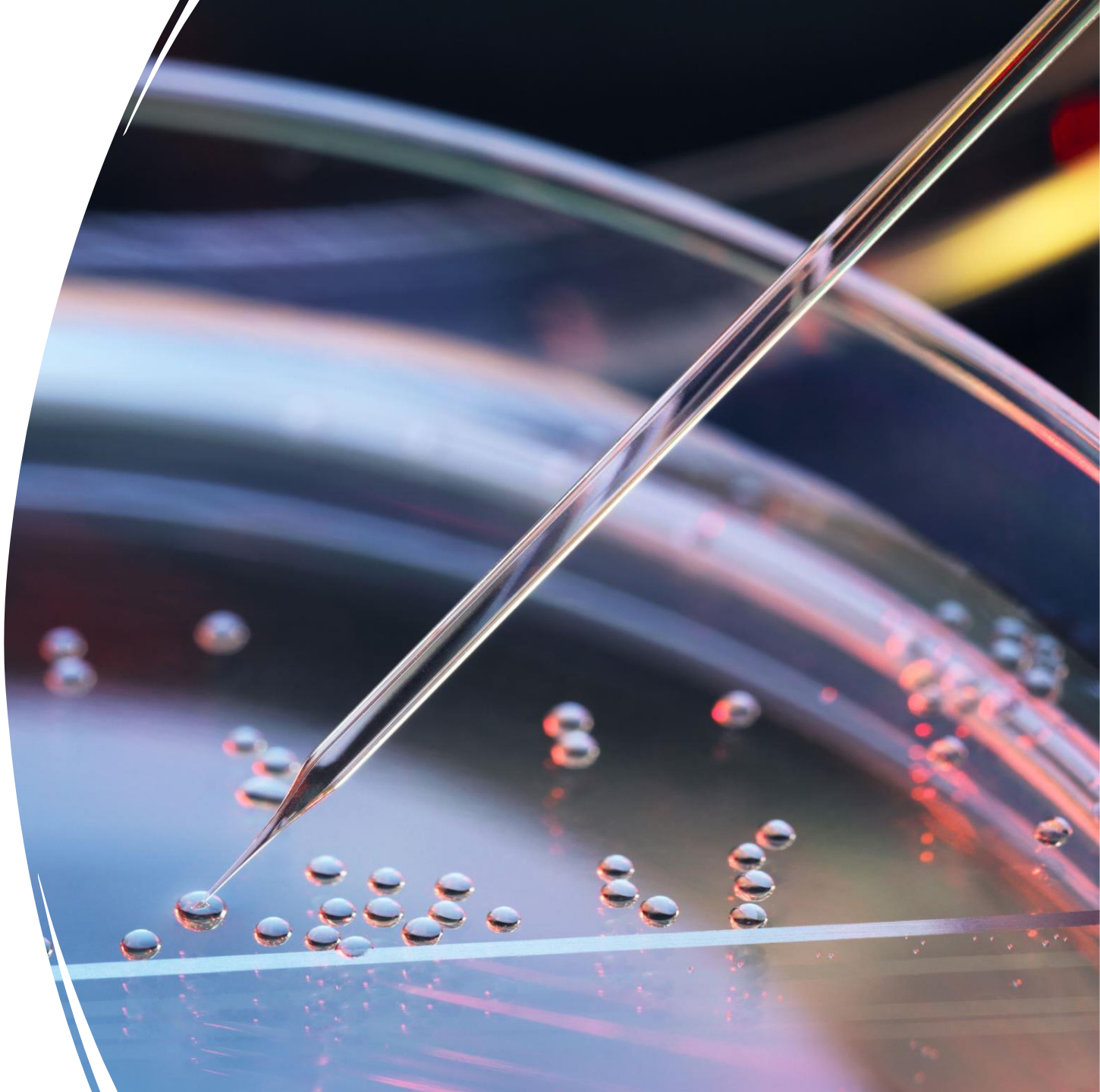
Physiological Effects of Ultraviolet :

- **1) Erythema due to UVR :**
- !! The redness caused by ultraviolet radiation is uniform, not mottled, and there is a sharp distinct edge at the junction with an unexposed area .
- !! This indicates that it is capillaries in the dermis that primarily dilate and not the arterioles.



Erythema

- Since UVB and C are largely absorbed in the epidermis it is considered that the reaction is due to the release of chemical substances.
- !! Histamine substances are released especially during the early development, certain prostagladins, which may be released from epidermal keratinocytes as well as prostacyclin from epidermal cells in the blood vessels of the dermis.



Why Ultraviolet (UV) and infrared (IR) heating produce different skin responses (erythema) ?





1. **Type of Radiation**:

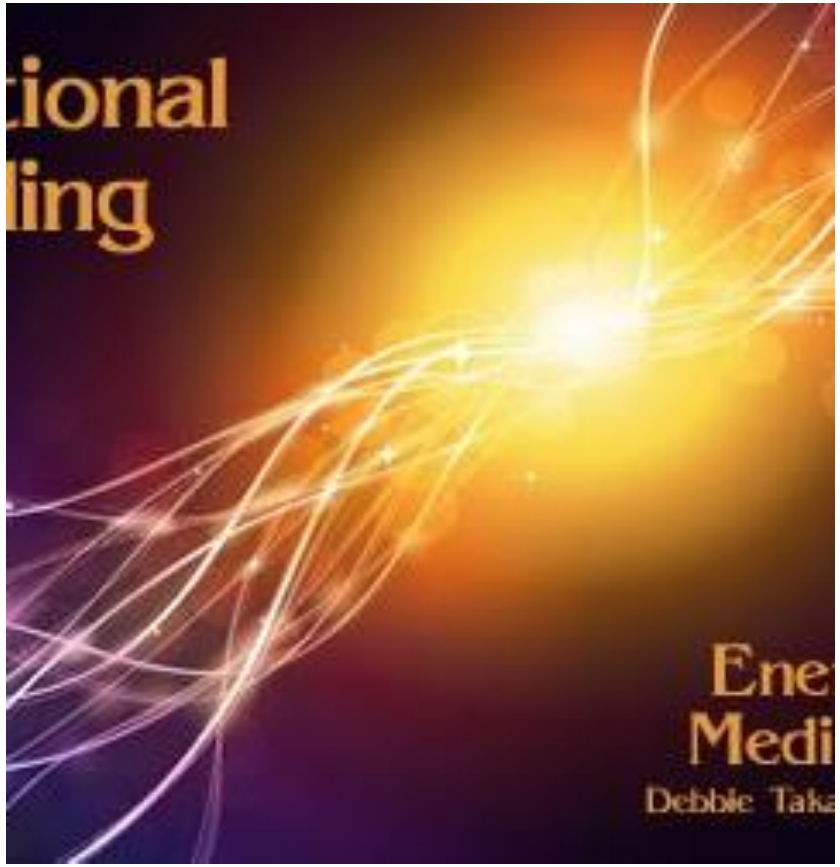
    1. **Ultraviolet Light**: UV radiation primarily affects the skin's outer layers (epidermis) and is known to cause damage to DNA in the cells. This damage leads to inflammation and erythema (redness) as a direct result of the biological reaction to this damage.

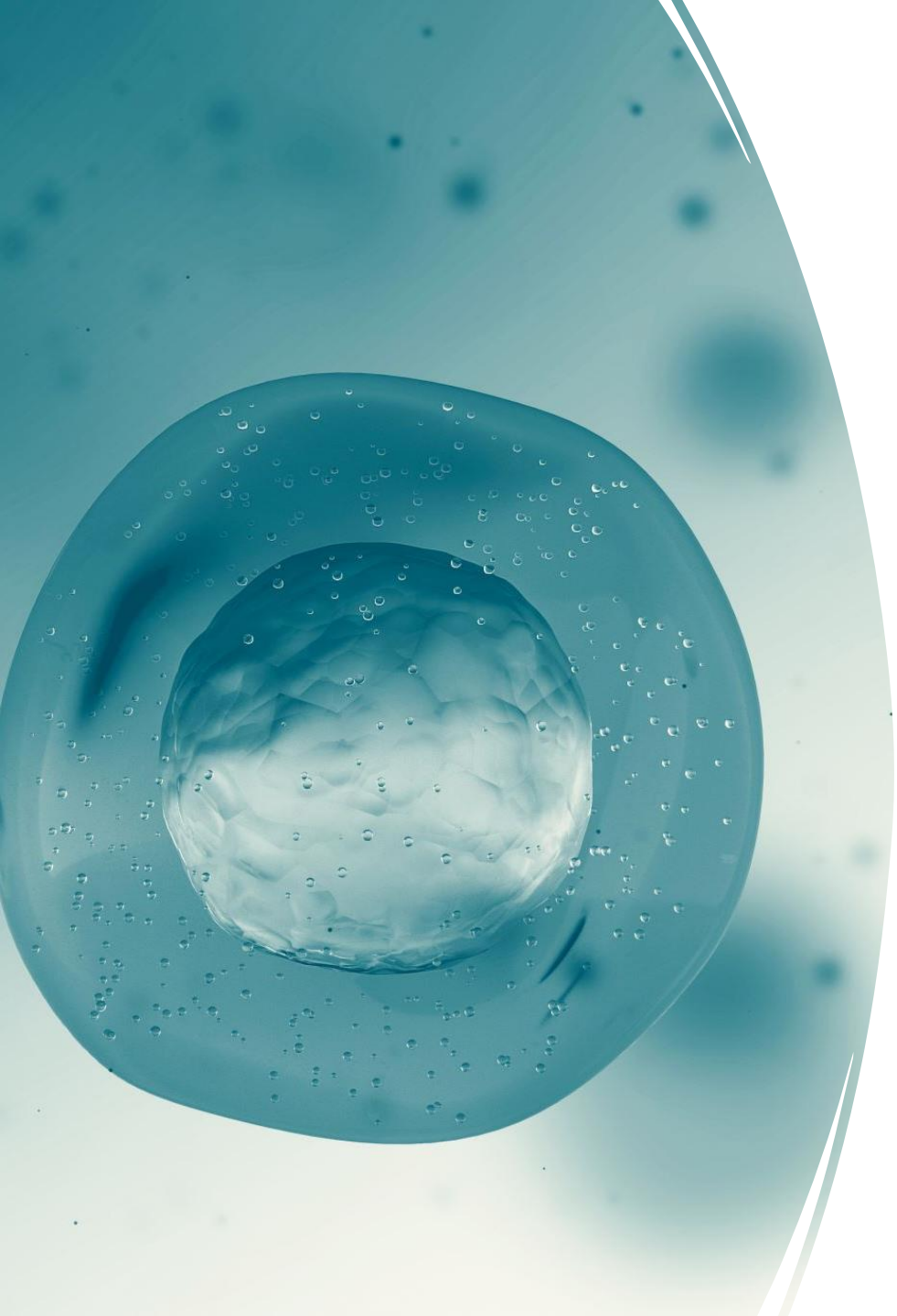
    2. **Infrared Light**: Infrared radiation penetrates deeper into the skin and primarily causes heating. It does not cause direct DNA damage like UV but instead raises the temperature of the tissue, leading to a uniform increase in blood flow.

2. **Skin Response**:

1. **Erythema (UV exposure)**: When the skin is exposed to UV light, especially UVB radiation, it induces a localized inflammatory response, resulting in patchy redness and swelling. This occurs because the immune response is activated due to cellular damage, which can lead to uneven or mottled appearances if the exposure varies across the skin.

2. **Uniform Heating (IR exposure)**: Infrared light warms up the tissue uniformly, which usually leads to a consistent and even erythema if any redness is produced. The heating can increase blood flow uniformly across the area exposed, thereby contributing to a more homogeneous appearance.





3. Mechanism of Action:

- Mottled Appearance (from UV): The mottled appearance can be due to varying degrees of individual skin response or uneven exposure to UV radiation (like in sunburn), where some areas may be more sensitive or were exposed longer than others.
- Uniform Appearance (from IR): Infrared exposure tends to be more evenly distributed across the surface area, leading to a more uniform appearance in terms of warmth and potential redness but not the same inflammatory response as with UV.

nema


Paediatr Cardiol


Rash on face
"Lacelike" rash
on extremities



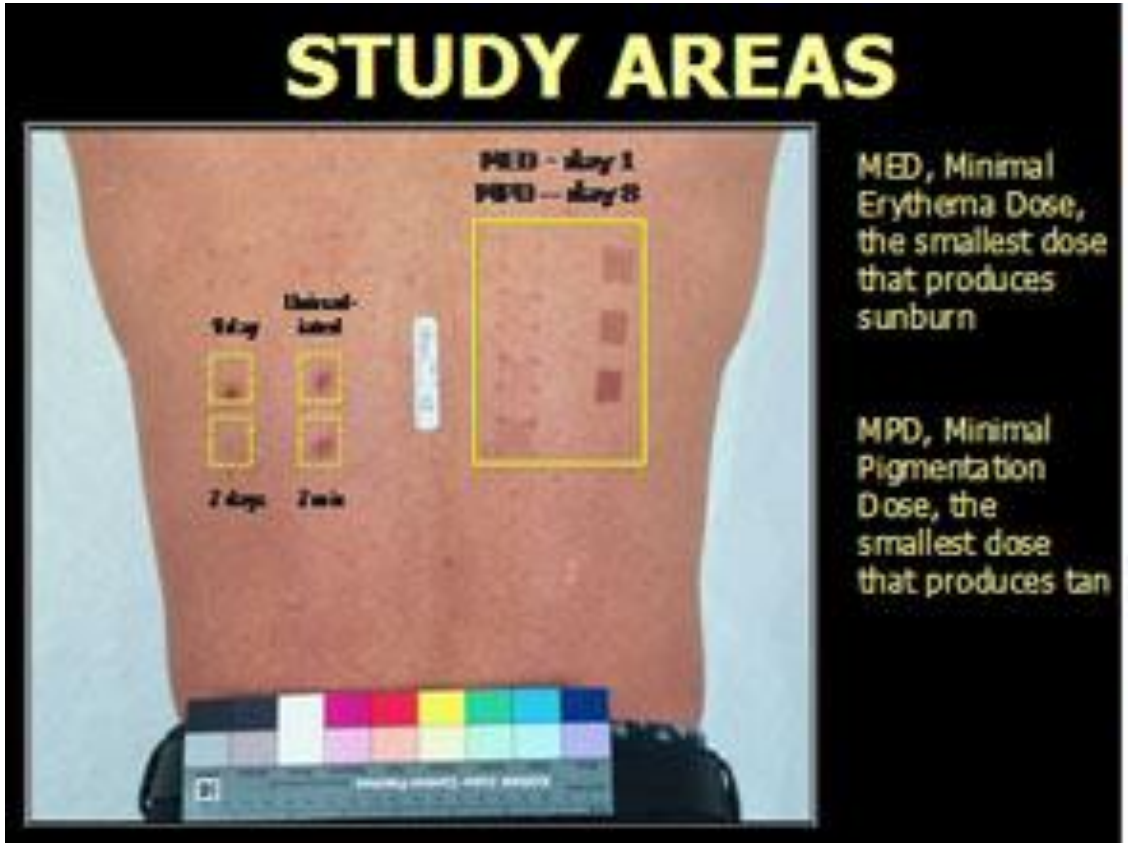


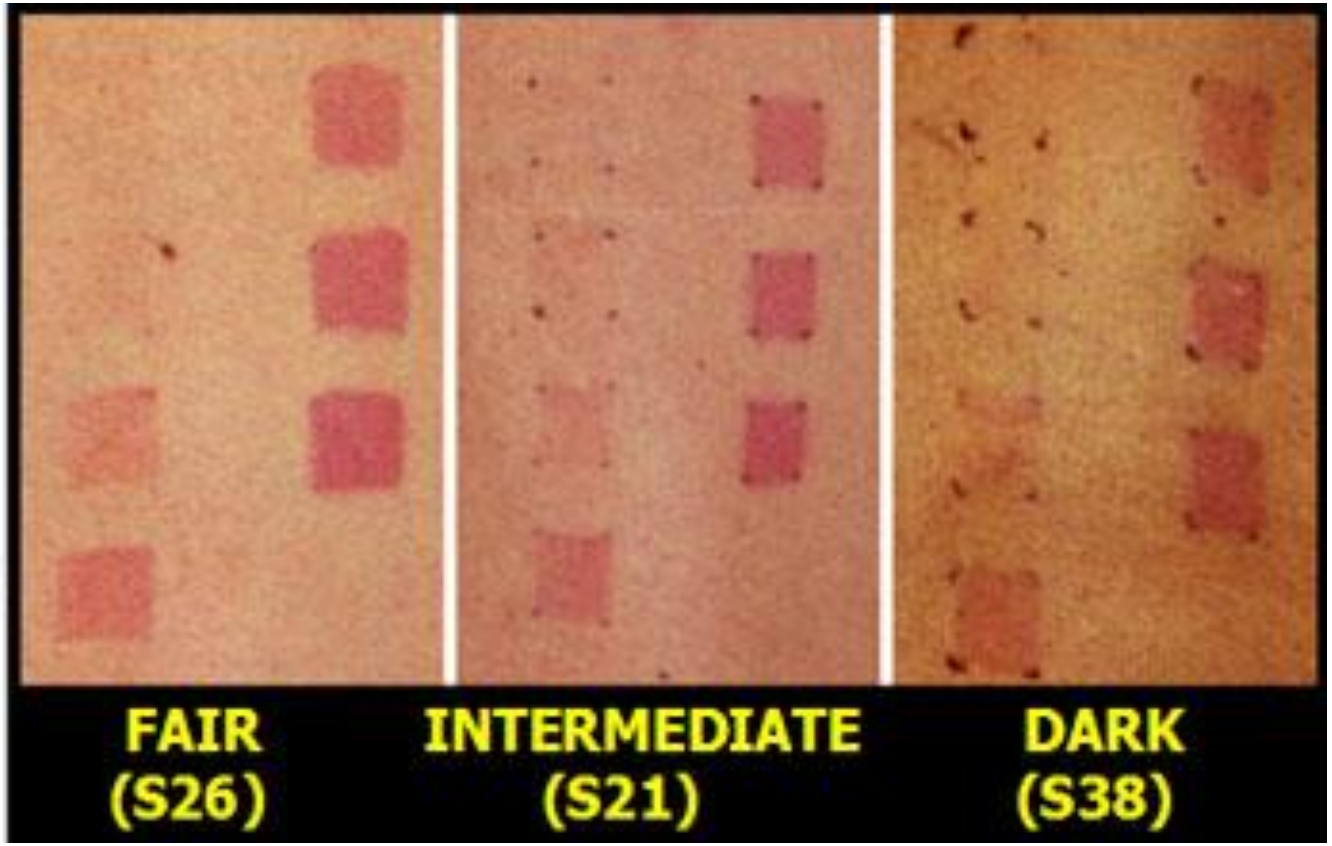
Mottled skin fashion in infrared

Degree of erythema

	Approximate latent period	Appearance	Approximate duration of erythema	Skin oedema	Skin discomfort	Desquamation of skin	Relation to dose causing E1
E1	**6 –12 hours**	Mildly pink	< 24 hours	None	None	None	1
E2	**6 hours**	Definite pink-red Blanches on pressure	2 days	None	Slight soreness Irritation	Powdery	2.5
E3	**3 hours**	Very red Does not blanch on pressure	3 – 5 days	Some	Hot and painful	In thin sheets	5
E4	**< 2 hours**	Angry red	Week	Blister	Very painful	Thick sheets	10

Degree of erythema

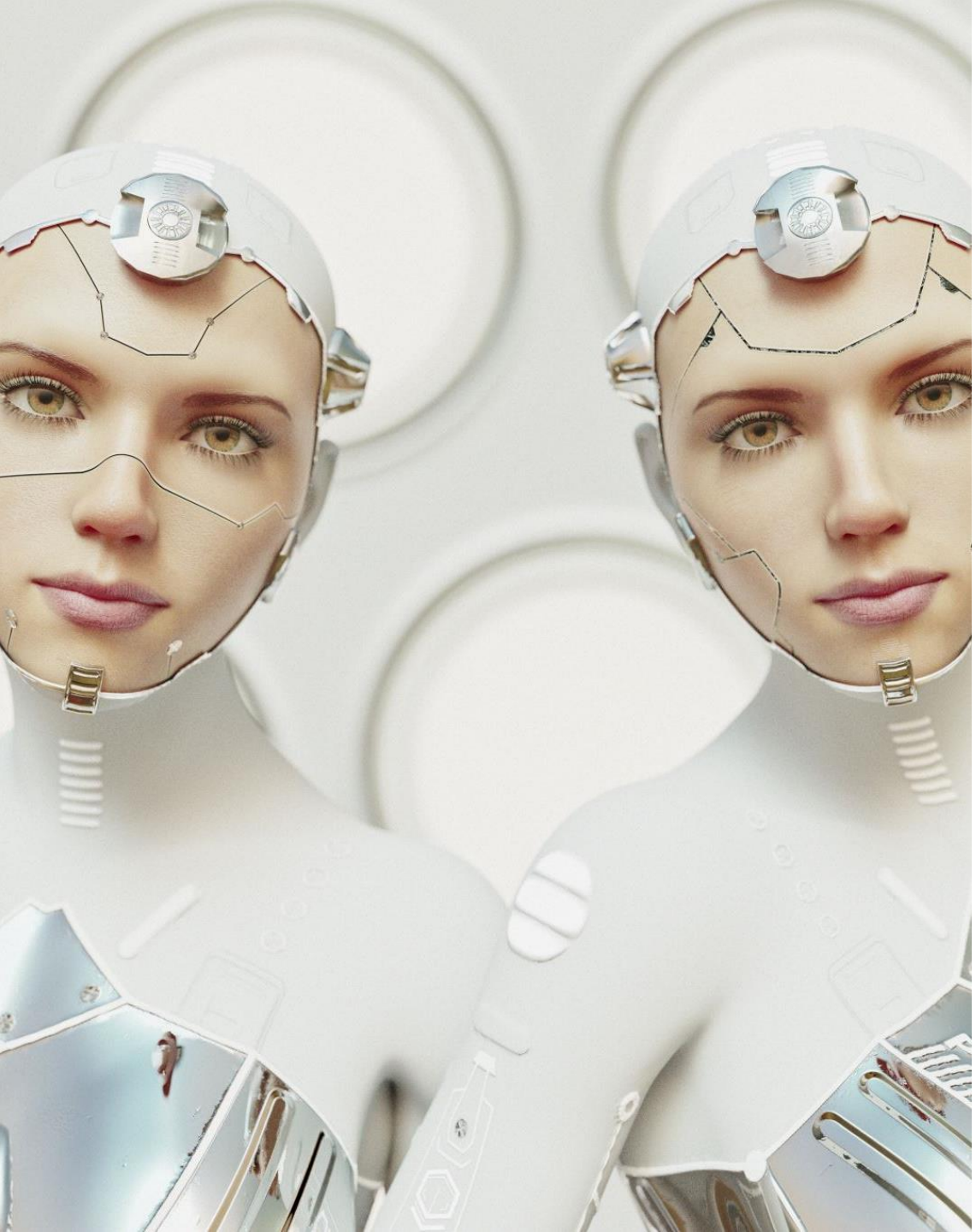






Minimal Erythemal Dose

- Visual inspection after 24 hours to determine MED
  - No erythema = suberythemal dose
- Erythema still present at 48 hours
  - 1st degree erythemal dose (2.5 times MED)
- Erythema persists from 48-72 hours
  - 2nd degree erythemal dose (5 times MED)
- Erythema lasting past 72 hours
  - 3rd degree erythemal dose



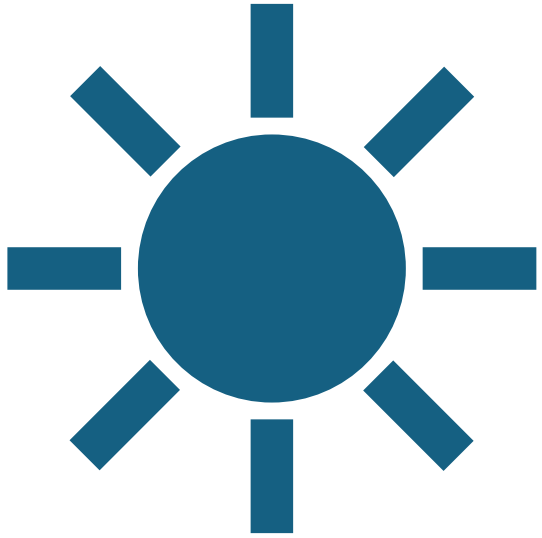
Minimal Erythemal Dose

- Skin adapts to UVR exposure, therefore, MED will gradually increase with repeated treatments
- Must gradually increase exposure time to achieve the same reaction
  - Increased by 5 seconds per treatment
- Height of lamp remains constant

Answer:

- One of important cutaneous vasodilator is nitric oxide, also contributes to the maintenance of erythema due to UVB.
- A gradual diffusion of this chemical takes place until sufficient has accumulated around the blood vessels in the skin to make them dilate. This accounts for the latency of the erythema.

Why UV erythema Take long time to appear than IR ?





Physiological Effects of Ultraviolet :

2) Pigmentation :

- Pigmentation of the skin occurs as a result of both the formation of melanin in the deep region of the epidermis and the migration of melanin already formed into more superficial layers.
- In already pigmented individuals some immediate tanning may occur within as little as 10 minutes of exposure, due to darkening of existing melanin by photo-oxidation.

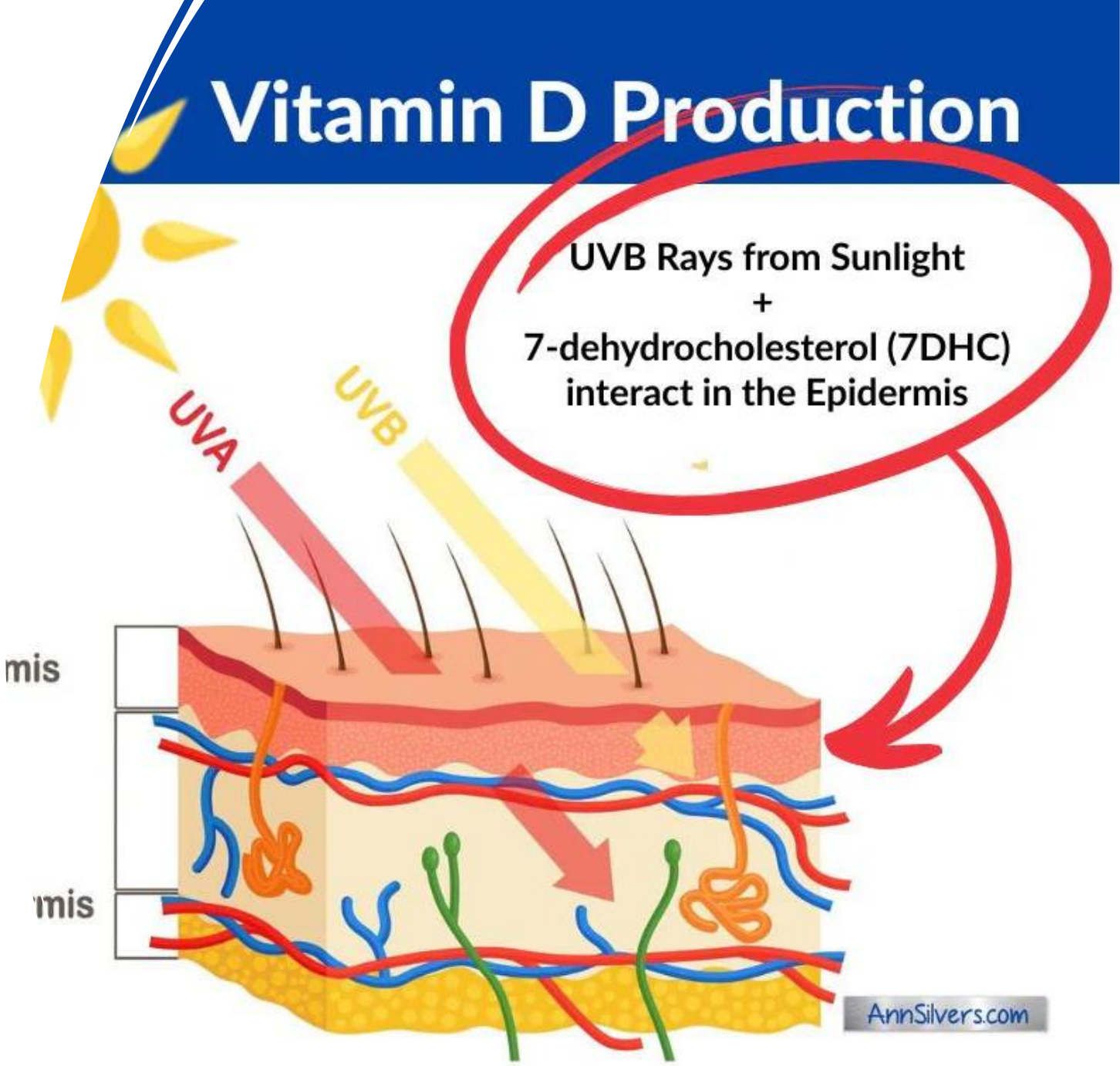


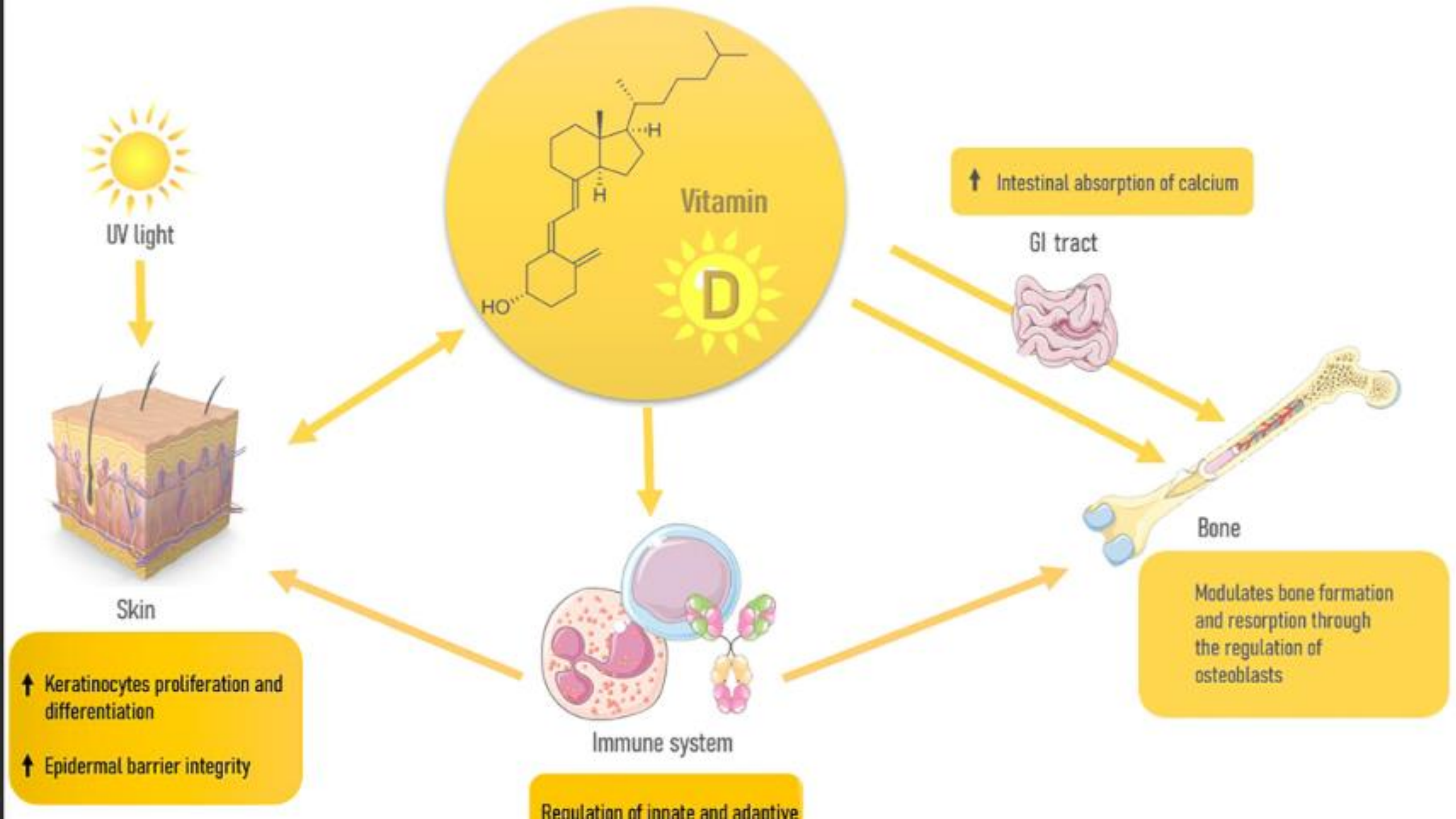


3) Increased Skin Growth

- Stimulation by ultraviolet radiation provokes increased keratinocyte cell turnover so that the skin grows more rapidly for a time, leading to shedding of the most superficial cells at an earlier stage in their development than usual so that they remain in pieces, or even sheets, and can be peeled off.

- Ultraviolet radiation also has positive effects on the human body, however. It stimulates the production of vitamin D in the skin and can be used as a therapeutic agent for such diseases as psoriasis. Because of its bactericidal capabilities at wavelengths of 260–280 nm, ultraviolet radiation is useful as both a research tool and a sterilizing technique. Fluorescent lamps exploit the ability of ultraviolet radiation to interact with materials known as phosphors that emit visible light; compared with incandescent lamps, fluorescent lamps are a more energy-efficient form of artificial lighting.




UV light
Vitamin
D
HO
H
H
Intestinal absorption of calcium
GI tract
Skin
Bone
Modulates bone formation and resorption through the regulation of osteoblasts
Keratinocytes proliferation and differentiation
Epidermal barrier integrity
Immune system
Regulation of innate and adaptive

4) Vitamin D Production :

- UVB is able to convert sterols in the skin, such as 7-dehydro-cholesterol to vitamin D which (after changes in the liver and kidneys) is able to facilitate the absorption of calcium from the intestine.

(**Sterols** are common components of human **skin** oils.)



After Sun Exposure, Be Careful about Showering!



- It`s important to understand that vitamin D3 is an oil soluble steroid hormone. It`s formed when your skin is exposed to ultraviolet B (UVB) radiation from the sun (or a safe tanning bed). When UVB strikes the surface of your skin, your skin converts a cholesterol derivative in your skin into vitamin D3.

Con.

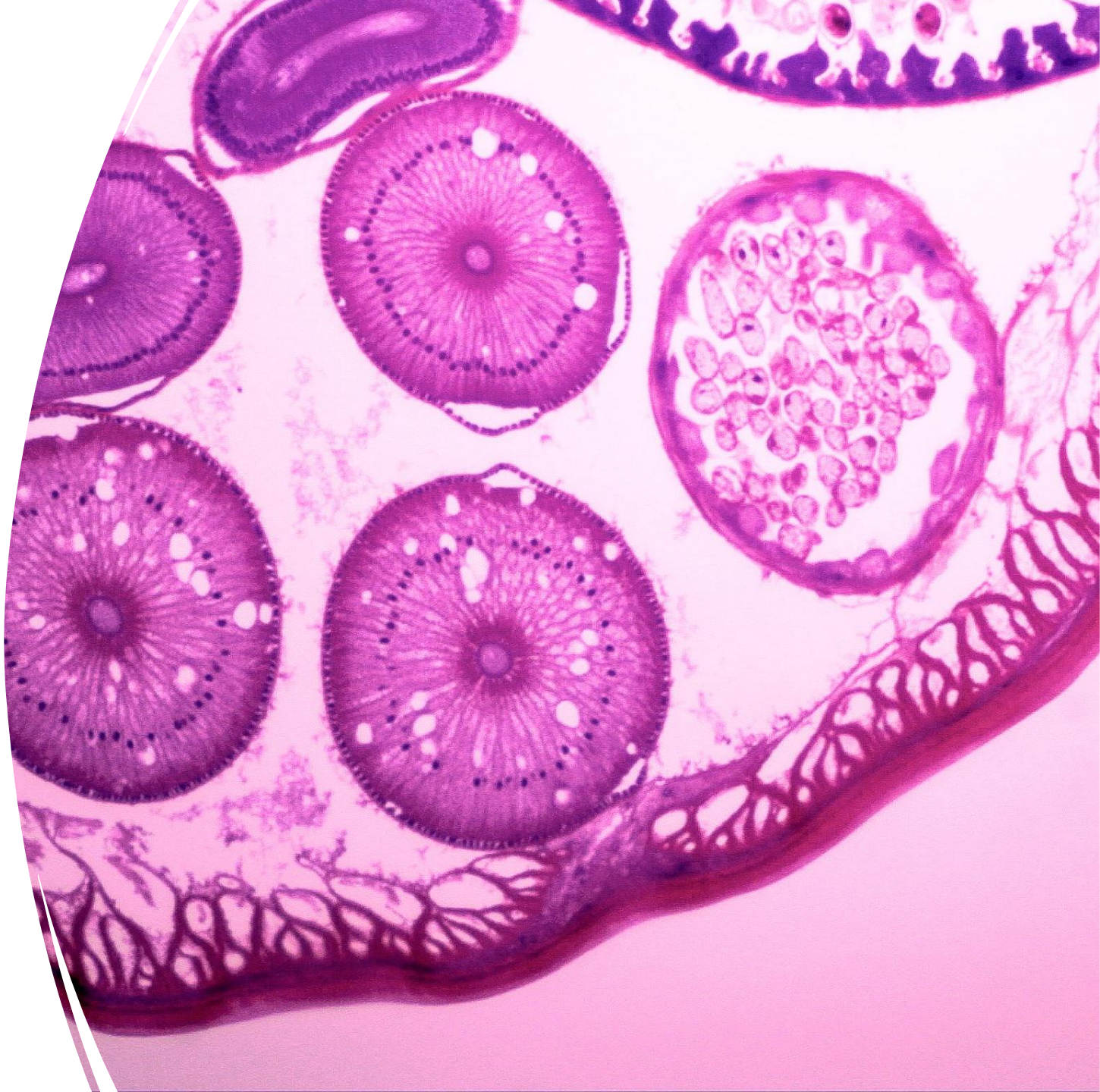
- However, the vitamin D3 that is formed is on the surface of your skin does not immediately penetrate into your bloodstream. It actually needs to be absorbed from the surface of your skin into your bloodstream.



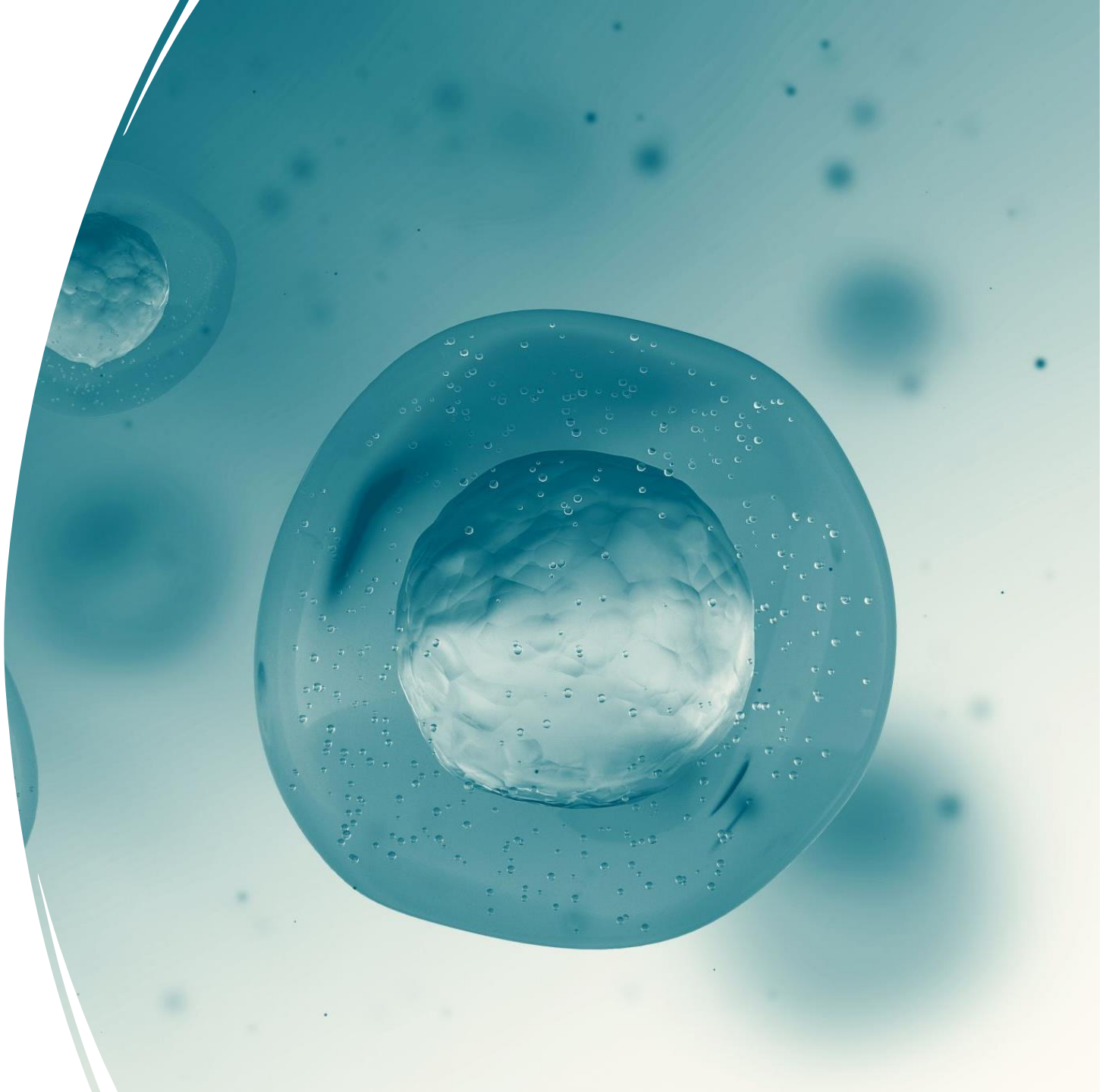
The critical question then is:
How long does it take the vitamin D3 to penetrate your skin and reach your bloodstream?

5) Immunosuppressive Effects :

- An ultraviolet radiation appears to trigger immunosuppressive effects, both locally and systemically. This occurs because UVB destroys Langerhan’s cells and stimulates the proliferation of suppressor T cells.



- **Langerhans** cells are dendritic cells (antigen-presenting immune cells) of the skin, and contain organelles called Birbeck granules. They are present in all layers of the epidermis and responsible for skin immunity



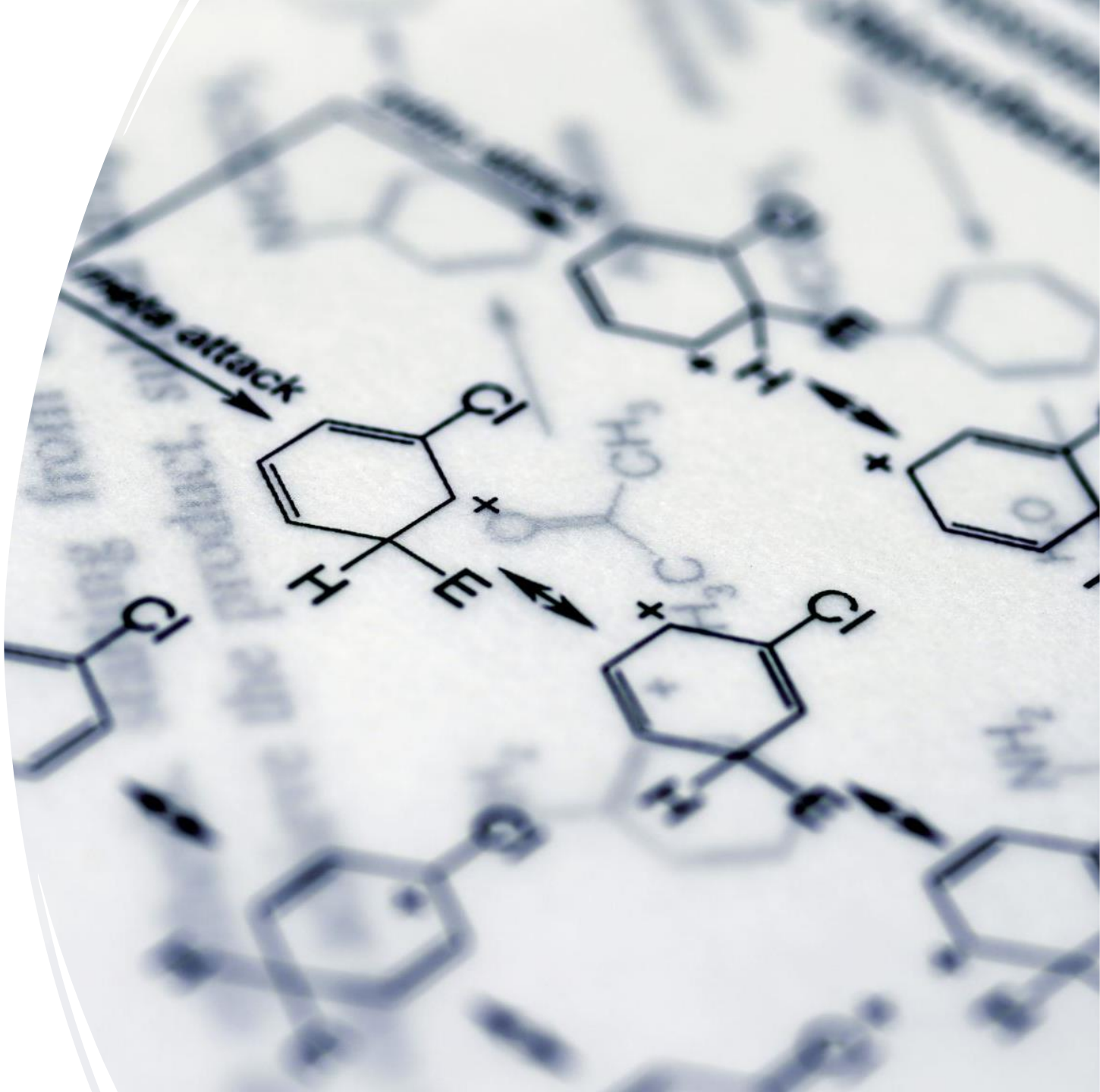
CON.

- When organisms invade the skin, the macrophage-like Langerhans cells gather some of the pathogen and transport it to the lymph nodes, which send out specific killer T cells.



Con.

- Suppressor T cells are regulatory in that they inhibit antibody production and suppress the action of other T cells. This immunosuppressive effect is believed to be the protective response to prevent an autoimmune attack on the skin cells that have been altered by ultraviolet radiation.



Effect on Cells



UVR causes chemical excitation of cells



Results in alteration of cell biochemistry and cellular metabolism

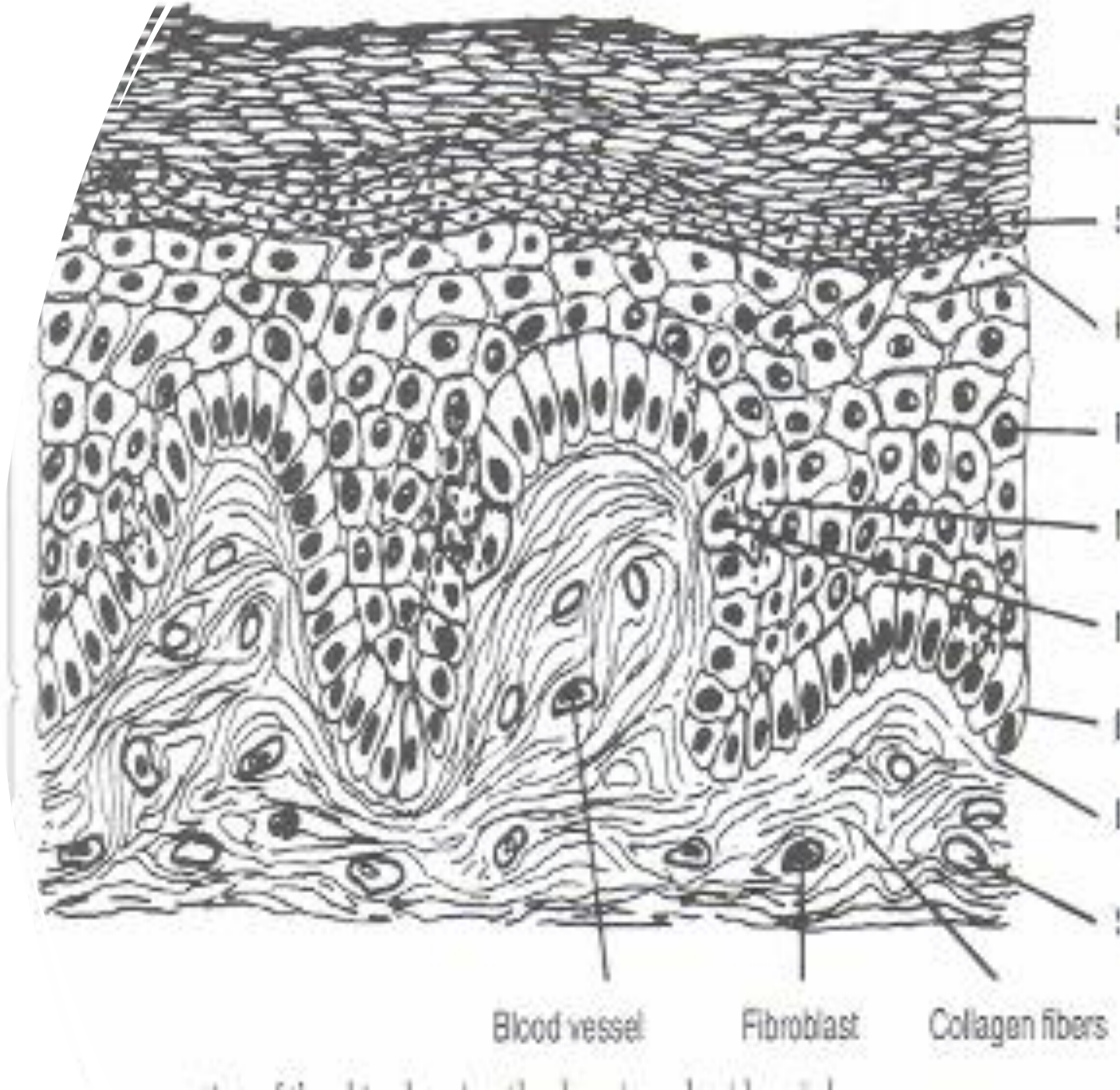
Affects synthesis of DNA and RNA



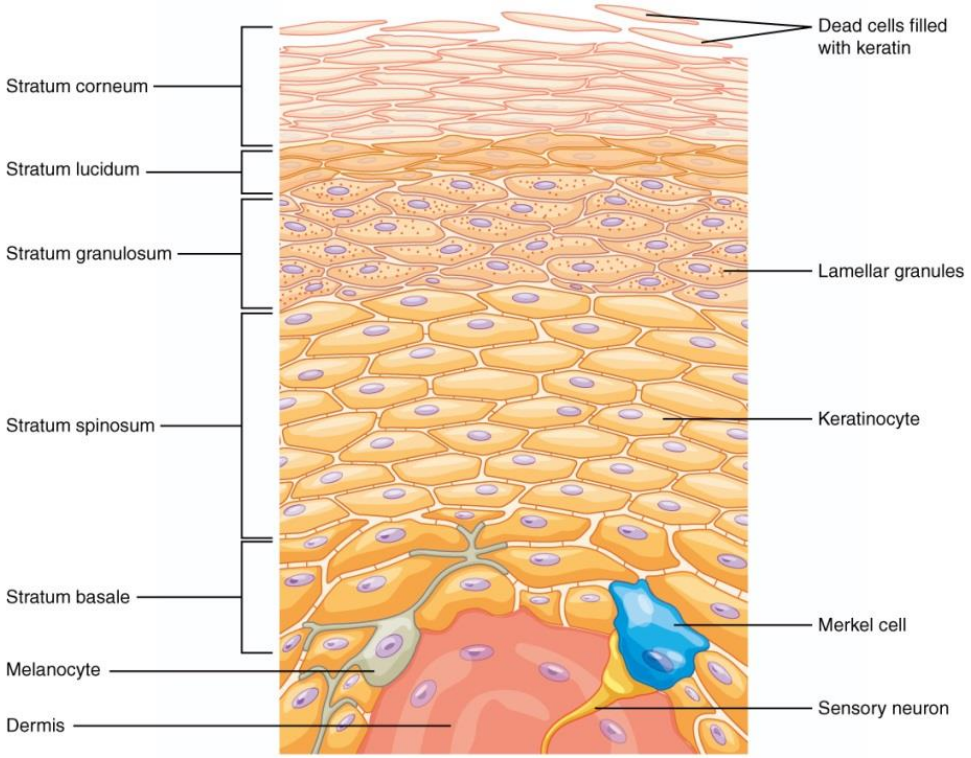
Protein and enzyme production is altered, which may result in cell death

Effects on Skin

- Epidermis
  - Keratinocytes, which provide fibrous protective protein of skin
- Dermis
  - Papillary layer - rich blood supply
  - Reticular layer - heavy connective tissue containing fibroblasts, histocytes, and mast cells



Desquamations





Effects on Skin: Erythema

Generalized response to UVR exposure	Acute inflammatory reaction	End results:
• Reddening of skin		• Erythema - sunburn • Pigmentation - tanning • Increased epidermal thickness



Effects on Skin: Photosensitization

- Process in which a person becomes overly sensitive to UVR
- Acute effects of UVR exposure may be aggravated if certain chemicals or medications are present on skin (or in body)

Effects on Skin: Tanning

- Increase of pigmentation
  - Protective mechanism activated by UVR exposure
- Increase of ***melanin***
  - Pigment responsible for darkening skin
- Melanin functions as a biologic filter of UVR
  - By scattering radiation
  - By absorbing UVR
  - By dissipating absorbed energy as heat



Effects on Skin: Long-term

- Premature aging of the skin
- Dryness, cracking, and decreased elasticity
  - Linked to UVR-induced DNA damage
- Skin cancer
  - Most common malignant tumor found in humans
    - Basal cell carcinoma (rarely metastasizes)
    - Squamous cell carcinoma (metastasizes in 5%)
    - Malignant melanoma (usually metastasizes)
  - Damage to DNA suspected as cause
  - Rate of cure exceeds 95% with early detection



Effects on Eyes

- UVR exposure causes acute inflammation called *photokeratitis*
- Delayed reaction, occurs within 6 - 24 hours
- Signs/Symptoms:
  - Conjunctivitis accompanied by erythema of adjacent facial skin
  - Sensation of a foreign body on eye
  - Photophobia
  - Increased tears production
  - Spasm of the ocular muscles



Systemic Effects

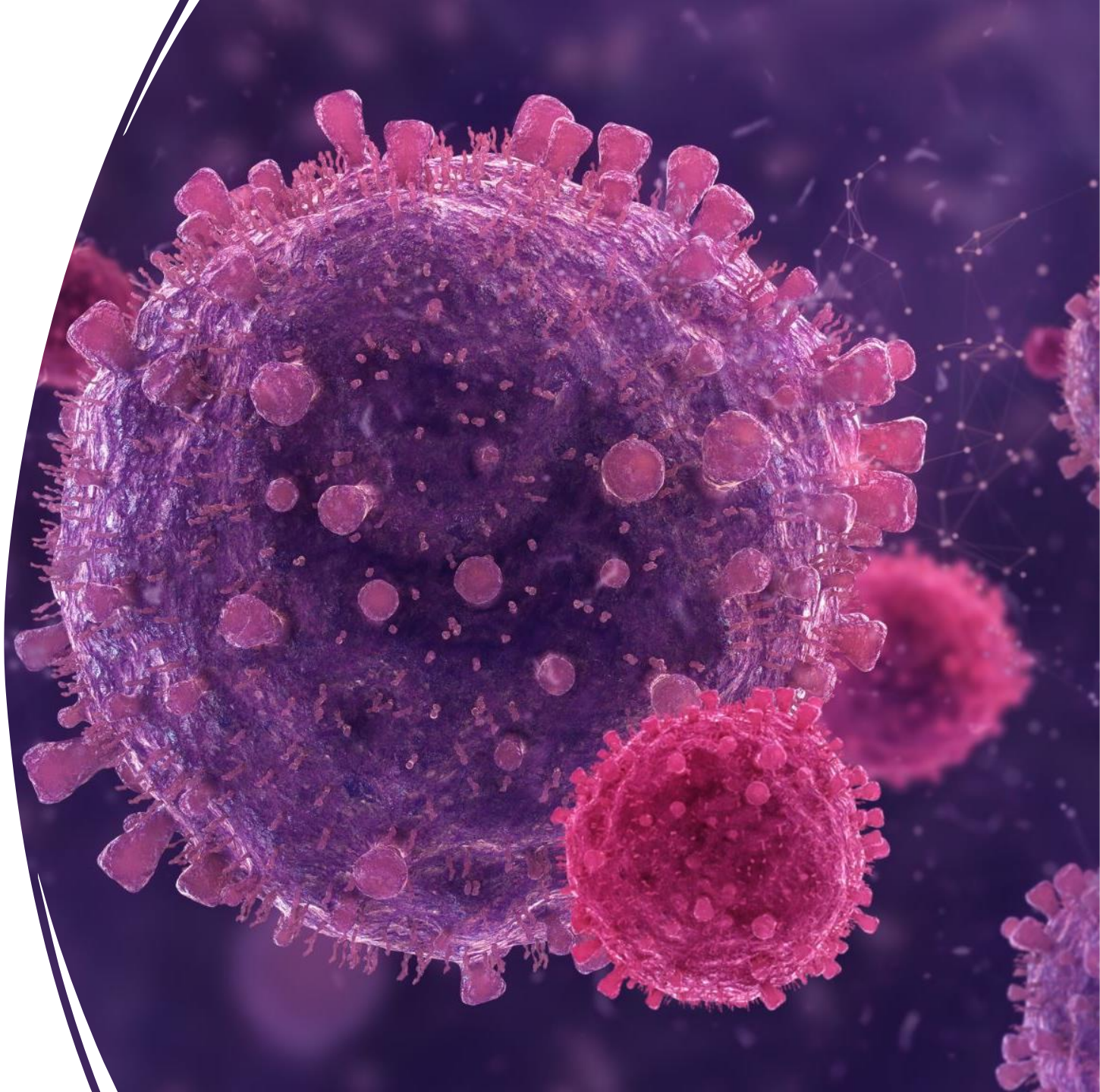
- Photosynthesis of vitamin D
  - Following exposure to UVR in UV-B range
- May be used to treat Ca^{++} and phosphorus disorders
  - Rickets and tetany
- Treatment of choice; however, is dietary supplementation



Clinical Applications

- Dermatologic conditions
  - Psoriasis, acne, and hard to cure infectious skin conditions such as pressure sores

- Development of oral and topical medications has greatly reduced the use of ultraviolet



summary

- Ultraviolet
  - Beneficial effects
    - Increases the elimination of waste products
    - Improves the flow of blood & lymph
    - Has a germicidal & antibacterial effect
    - Produces vita min D in the skin
    - Can be used to treat rickets (soft & deformed bones), psoriasis & acne
    - Produces a tan

Indications

- Acne
- Aseptic wounds
- Septic wounds
- Sinusitis
- Psoriasis

Pressure sores

Osteomalacia

Diagnosis of skin disorders

Increased vitamin D production

Sterilization

Tanning

Hyperplasia

Contraindications

- Acute psoriasis
- Acute eczema
- Herpes simplex
-
-
-

Renal and hepatic insufficiencies

Diabetes

Hyperthyroidism

Mercury Arc Lamp

- Positioning
  - Apply cosine law and inverse square law
- Distance of lamp must be kept constant if intensity of treatments is to be equal
- Standardized at each clinic
  - Usually ranges between 24 - 40 inches





Technique of applications

